6/4

a)

$f(x) = \arcsin(x)$

$f'(x) = \frac{1}{\sqrt{1-x^2}}$

$f''(x) = \left((1-x^2)^{-\frac{1}{2}}\right)' = -\frac{1}{2}(1-x^2)^{-\frac{3}{2}} \cdot (-2x) = x \cdot (1-x^2)^{-\frac{3}{2}}$

$f^{(3)}(x) = \left(x(1-x^2)^{-\frac{3}{2}}\right)' = 1 \cdot (1-x^2)^{-\frac{3}{2}} + x \cdot -\frac{3}{2}(1-x^2)^{-\frac{5}{2}} \cdot (-2x) = (1-x^2)^{-\frac{3}{2}} + 3x^2(1-x^2)^{-\frac{5}{2}} =$

$= \frac{2x^2+1}{(1-x^2)^{5/2}}$

$f\left(\frac{\sqrt{3}}{2}\right) = \arcsin\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{3}\pi$

$f'\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{\sqrt{1-\left(\frac{3}{4}\right)}} = \frac{1}{\sqrt{\frac{1}{4}}} = \frac{1}{\frac{1}{2}} = 2$

$f''\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{2} \cdot \left(1-\frac{3}{4}\right)^{-\frac{3}{2}} = \frac{\sqrt{3}}{2} \cdot \left(\frac{1}{4}\right)^{-\frac{3}{2}} = \frac{\frac{\sqrt{3}}{2}}{\left(\frac{1}{4}\right)^{\frac{3}{2}}} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{8}} = 4\sqrt{3}$

$f^{(3)}\left(\frac{\sqrt{3}}{2}\right) = \frac{\frac{6}{4}+1}{\left(\frac{1}{4}\right)^{\frac{5}{2}}} = \frac{\frac{10}{4}}{\frac{1}{32}} = \frac{320}{4} = 80$

$T_3(x) = \frac{\pi}{3} + 2\left(x-\frac{\sqrt{3}}{2}\right) + \frac{4\sqrt{3}}{2}\left(x-\frac{\sqrt{3}}{2}\right)^2 + \frac{80}{6}\left(x-\frac{\sqrt{3}}{2}\right)^3 = \underline{\underline{\frac{\pi}{3} + 2\left(x-\frac{\sqrt{3}}{2}\right) + 2\sqrt{3}\left(x-\frac{\sqrt{3}}{2}\right)^2 + \frac{40}{3}\left(x-\frac{\sqrt{3}}{2}\right)^3}}$

7/1 - c)

$f'(x) = 0$

$(|(x-1)^3 - 27| - 8)' = 0$

$(x-1)^3 - 27 = 0$

$(x-1)^3 = 27$

$x - 1 = 3$

$x = 4$

$x \leq 4$

$(-(x-1)^3 + 27 - 8)' = 0$

$(-(x-1)^3 + 19)' = 0$

$-3(x-1)^2 \cdot 1 = 0$

$-3(x-1)^2 = 0$

$(x-1)^2 = 0$

$x = 1$

$x \geq 4$

$((x-1)^3 - 27 - 8)' = 0$

$((x-1)^3 - 35)' = 0$

$3(x-1)^2 = 0$

$(x-1)^2 = 0$

$x = 1$

$\downarrow$

není na intervalu

$f'(2) = -3(2-1)^2 = -3(1)^2 = -3$

$f'(0) = -3(0-1)^2 = -3 \cdot (-1)^2 = -3$

$f'(5) = 3(5-1)^2 = 3 \cdot (4)^2 = 48$

$f$ je klesající na intervalu $(-\infty; 4\rangle$ se stacionárním bodem $x = 1$ a rostoucí na intervalu $\langle 4; \infty)$

7/2 - b)

i)

$(\sin x + \cos^2 x)' = 0$

$\cos x + 2\cdot(\cos x)\cdot(-\sin x) = 0$

$\cos x - 2\cos x \sin x = 0$

$(-\cos x)\cdot(2\sin(x) - 1) = 0$

$-\cos x = 0$

$x \in \{\frac{\pi}{2}, \frac{3\pi}{2}\}$

$2\sin(x) - 1 = 0$

$\sin(x) = \frac{1}{2}$

$x \in \{\frac{\pi}{6}; \frac{5\pi}{6}\}$

$f(0) = \sin(0) + \cos^2(0) = 0 + 1^2 = 1$

$f(\frac{\pi}{2}) = \sin(\frac{\pi}{2}) + \cos^2(\frac{\pi}{2}) = 1 + 0^2 = 1$

$f(\frac{3\pi}{2}) = \sin(\frac{3\pi}{2}) + \cos^2(\frac{3\pi}{2}) = -1 + 0^2 = -1$

$f(\frac{\pi}{6}) = \sin(\frac{\pi}{6}) + \cos^2(\frac{\pi}{6}) = \frac{1}{2} + \frac{3}{4} = \frac{5}{4}$

$f(\frac{5\pi}{6}) = \sin(\frac{5\pi}{6}) + \cos^2(\frac{5\pi}{6}) = \frac{1}{2} + \frac{3}{4} = \frac{5}{4}$

$f(2\pi) = \sin(2\pi) + \cos^2(2\pi) = 0 + 1^2 = 1$

$\Rightarrow$ maximum $f(\frac{\pi}{6}) = f(\frac{5\pi}{6}) = \frac{5}{4}$

minimum $f(\frac{3\pi}{2}) = -1$

ii)

$\lim_{x \to 0^+} f(x) = 1$

$f(\frac{\pi}{2}) = 1$

$f(\frac{5\pi}{6}) = f(\frac{\pi}{6}) = \frac{5}{4}$

$\lim_{x \to \pi^-} f(x) = 1$

$\Rightarrow$ maximum $f(\frac{\pi}{6}) = f(\frac{5\pi}{6}) = \frac{5}{4}$

minimum $f(\frac{\pi}{2}) = 1$

iii)

z minulých částí plyne že maxima nenabývá, minimum $f(\frac{3\pi}{2}) = -1$

7/4 f-c)

$D(f) = (0;1) \cup (1;\infty)$

pro $x \in (0;1)$

$$(\ln(-\ln(x)))' = 0$$

$$\frac{1}{-\ln(x)} \cdot \frac{-1}{x} = 0$$

pro $x \geq 1$

$$(\ln(\ln(x)))' = 0$$

$$\frac{1}{\ln(x)} \cdot \frac{1}{x} = 0$$

$$f''(x) = \left(\frac{1}{\ln(x)} \cdot \frac{1}{x}\right)' = -\ln(x)^{-2} \cdot \frac{1}{x} \cdot \frac{1}{x} + \frac{1}{\ln(x)} \cdot -x^{-2} = -\frac{\ln(x)+1}{x^2 \ln^2(x)}$$

$$-\frac{\ln(x)+1}{x^2 \ln^2(x)} = 0$$

$$\ln(x)+1 = 0$$

$$\ln(x) = -1$$

$$x = \frac{1}{e}$$

$$f''(0{,}2) = -\frac{\ln(0{,}2)+1}{(0{,}2)^2 \ln^2(0{,}2)} = -\frac{-0{,}61}{\frac{1}{25} \cdot 2{,}54} \Rightarrow f''(0{,}2) > 0$$

$$f''(0{,}5) = -\frac{\ln(0{,}5)+1}{(0{,}5)^2 \cdot \ln^2(0{,}5)} = -\frac{0{,}31}{(0{,}5)^2 \cdot \ln^2(0{,}5)} \Rightarrow f''(0{,}5) < 0$$

$$f''(2) = -\frac{\ln(2)+1}{2^2 \cdot \ln^2(2)} = -\frac{1{,}69}{4 \ln^2(2)} \Rightarrow f''(2) < 0$$

f je konvexní na $(0; \frac{1}{e}\rangle$ a konkávní na $\langle \frac{1}{e}; 1) \cup (1; \infty)$

7/5 - c)

$\sqrt{x^2+x-6} \geq 0$

$(x-2)\cdot(x+3) \geq 0$

| | $-3$ | $2$ | |
|---|---|---|---|
| $(x-2)$ | $-$ | $-$ | $+$ |
| $(x+3)$ | $-$ | $+$ | $+$ |
| | $+$ | $-$ | $+$ |

šitme

$y = x+1 \vee \infty$

$y = -3x-1 \vee -\infty$

$D(f) = (-\infty; -3\rangle \cup (2; \infty)$

$$\lim_{x\to-\infty^+} f(x) = \lim_{x\to-\infty^+} \left(2\sqrt{x^2+x-6} - x\right) = \langle\langle \sqrt{\infty^2} + \infty \rangle\rangle = \infty$$

$$\lim_{x\to-\infty^+} \frac{f(x)}{x} = \lim_{x\to-\infty^+} \left(\frac{2}{x}\sqrt{x^2+x-6} - \frac{x}{x}\right) = \langle\langle 0 + \frac{\infty}{-\infty} \rangle\rangle = \lim_{x\to-\infty^+} \frac{2\sqrt{x^2+x-6}}{x} - 1 =$$

$$= \lim_{x\to-\infty^+} -1 + 2\cdot\frac{\sqrt{x^2+x-6}}{x} = \lim_{x\to-\infty^+} -1 + 2\sqrt{\frac{x^2+x-6}{x^2}} = -3 = k$$

$$\lim_{x\to-\infty^+} f(x) - kx = \lim_{x\to-\infty^+} \left(2\sqrt{x^2+x-6} - x + 3x\right) = \lim_{x\to-\infty^+} 2\cdot\left(\sqrt{x^2+x-6} + x\right) = 2\lim_{x\to-\infty^+} \frac{6-x}{\sqrt{x^2+x-6}-x} =$$

$$= 2\lim_{x\to-\infty^+} \frac{6-x}{x\left(1-\frac{\sqrt{x^2+x-6}}{x}\right)} = 2\lim_{x\to-\infty^+} \frac{\overset{\to 0}{6}-1}{1} \lim_{x\to-\infty} \frac{1}{1-\frac{\sqrt{x^2+x-6}}{x}} = 2\cdot -1 \cdot \frac{1}{2} = -1 = q$$

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$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \left(2\sqrt{x^2+x-6} - x\right) = \infty$$

$$\lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} -1 + 2\sqrt{\frac{x^2+x-6}{x^2}} = \lim_{x\to\infty} -1 + 2\sqrt{\frac{1+\overset{0}{\frac{1}{x}}-\frac{6}{x^2}}{1}} = \lim_{x\to\infty} -1+2 = 1 = k$$

$$\lim_{x\to\infty} f(x) - kx = \lim_{x\to\infty} \left(2\sqrt{x^2+x-6} - x\right) - x = \lim_{x\to\infty} 2\sqrt{x^2+x-6} - 2x = 2\lim_{x\to\infty} \frac{x-6}{x+\sqrt{x^2+x-6}} =$$

$$= 2\lim_{x\to\infty} \frac{x-6}{x}\cdot\frac{1}{\frac{\sqrt{x^2+x-6}}{x}+1} = 2\cdot\frac{1}{2} = 1$$

7/5-e)

$$\frac{1}{\ln x} \in \langle -1; 1 \rangle$$

$$\Downarrow$$

$\ln x$ $\cancel{x \in (0;1) \cup (1;\infty)}$

$$x \in (0; \tfrac{1}{e}\rangle \cup (e; \infty)$$

$$\lim_{x\to\infty} \arccos\left(\frac{1}{\ln x}\right) = \arccos\left(\lim_{x\to\infty} \frac{1}{\ln x}\right) = \arccos(0) = \frac{\pi}{2}$$

$$\lim_{x\to\infty} \frac{\arccos\left(\frac{1}{\ln x}\right)}{x} = \left\langle\left\langle \frac{\frac{\pi}{2}}{\infty} \right\rangle\right\rangle = 0$$

vodorovná $y = \frac{\pi}{2}$

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8/2 e)

$$\int \frac{x^2+3}{5+5x^2}\,dx = \int \frac{1}{5}\,\frac{x^2+3}{1+x^2}\,dx = \int \frac{1}{5}\left(1+\frac{2}{1+x^2}\right)dx = \frac{1}{5}\left(x + 2\,\mathrm{arctg}(x)\right) + C$$